

Project Title: Smart Energy Consumption Analysis and Prediction using Machine Learning with Device-Level Insights

Milestone 2: Week 3-4

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Objective: Baseline Model Development

In this work, I import the **LinearRegression** class from the `sklearn.linear_model` module to build a regression-based prediction model. Linear Regression is used to model the relationship between the input features and the target variable, *Energy Consumption (kWh)*, by fitting a linear equation to the observed data.

First, I **define the file path** where the pre-processed, appliance-wise datasets are stored. Then, I **list all the files** present in this directory to identify the available datasets.

Next, I **extract the appliance names** from the filenames by selecting only those files that start with `x_test` and removing the unnecessary parts of the filename. I sort these appliance names and store them in a list so that the models can be trained separately for each appliance.

Evaluation, and Result Storage

After that, I **load the training data of a sample appliance (TV)** to verify the structure of the dataset. I check the data types and shapes of the feature matrix and target vector to ensure that the data is correctly loaded and suitable for model training.

Then, I **initialize an empty list** to store the performance results of each appliance-specific model. Using a loop, I **iterate through each appliance type** and load its corresponding training and testing datasets for both input features and target values.

Next, I **train a Linear Regression model** using the training data of the current appliance. Once the model is trained, I **generate predictions** on the test dataset.

After prediction, I **evaluate the model performance** using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These evaluation metrics help measure the prediction accuracy and error magnitude. The calculated results are stored in a list for later comparison across appliances.

Then, I **visualize the model performance** by plotting the actual and predicted energy consumption values against time steps. This graphical comparison helps in understanding how well the model follows the real consumption pattern.

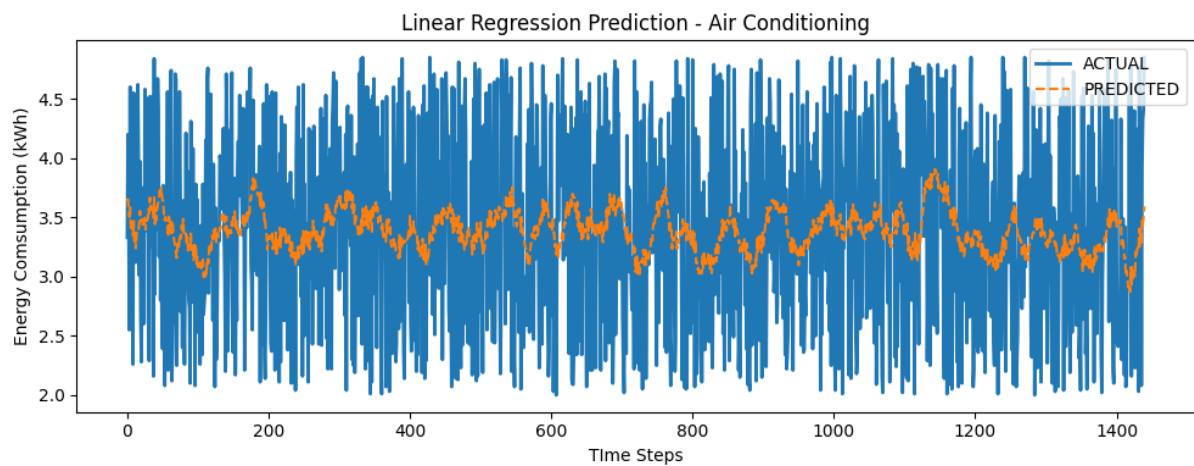
After visualization, I **save the trained model** for each appliance as a serialized `.pkl` file using `joblib`. This allows the trained models to be reused later without retraining.

Finally, I convert the collected evaluation results into a **DataFrame**, which provides a clear summary of model performance for all appliances.

Training Linear Regression for: Air Conditioning

MAE : 0.6830

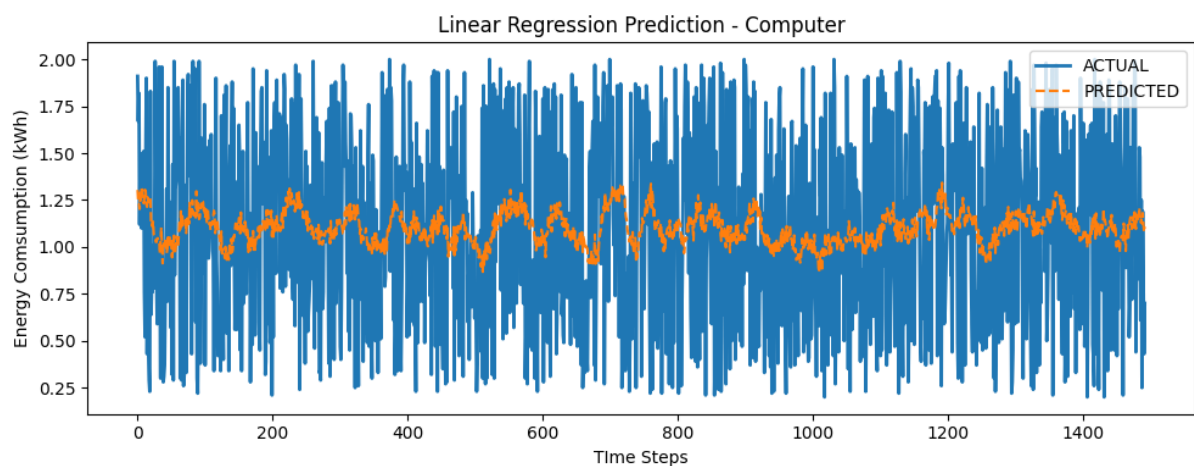
RMSE : 0.7999



Training Linear Regression for: Computer

MAE : 0.4363

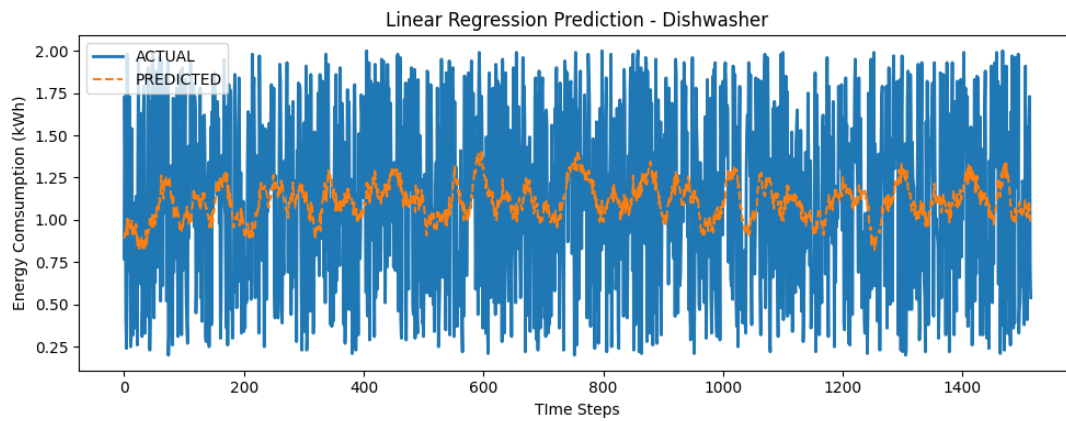
RMSE : 0.5037



Training Linear Regression for: Dishwasher

MAE : 0.4414

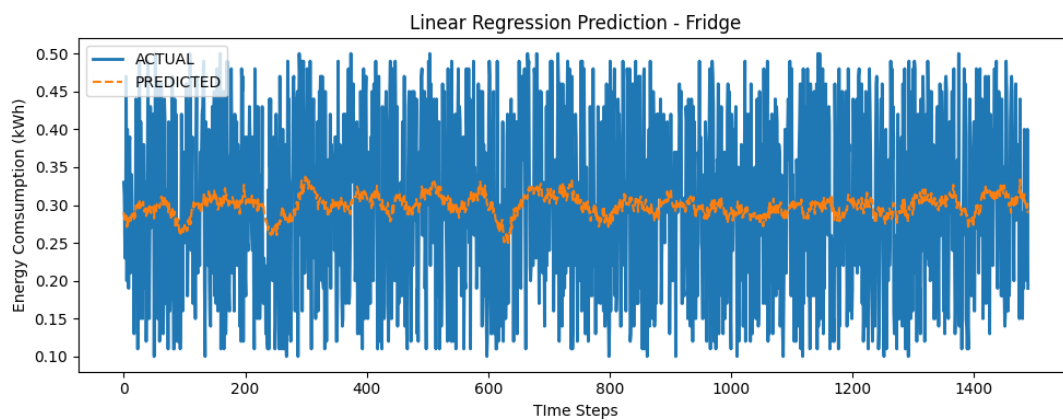
RMSE : 0.5137



Training Linear Regression for: Fridge

MAE : 0.0985

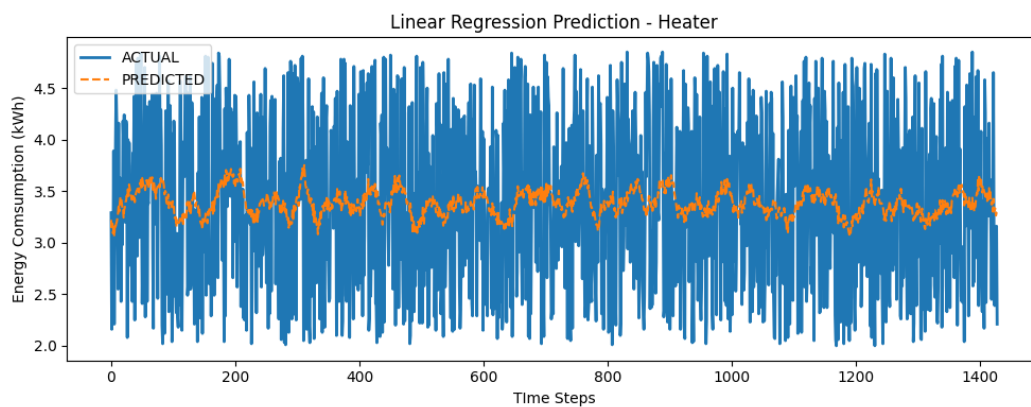
RMSE : 0.1137



Training Linear Regression for: Heater

MAE : 0.6955

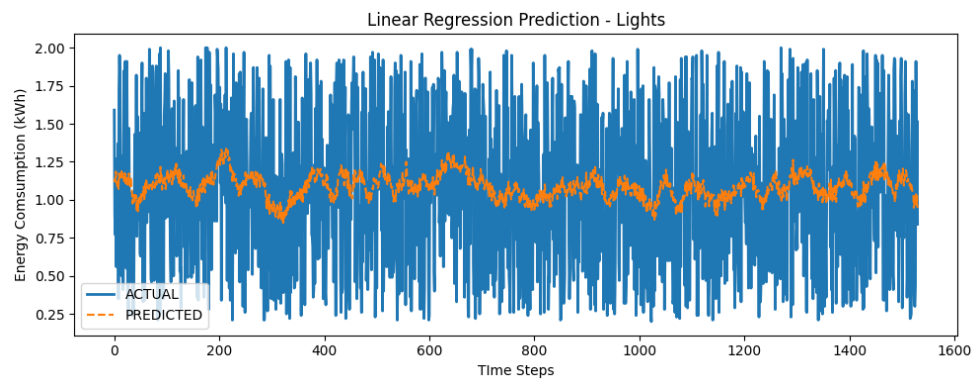
RMSE : 0.8049



Training Linear Regression for: Lights

MAE : 0.4289

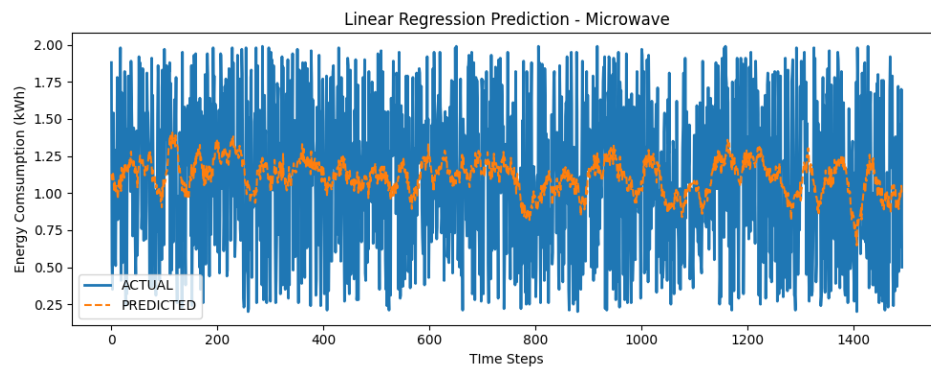
RMSE : 0.4995



Training Linear Regression for: Microwave

MAE : 0.4331

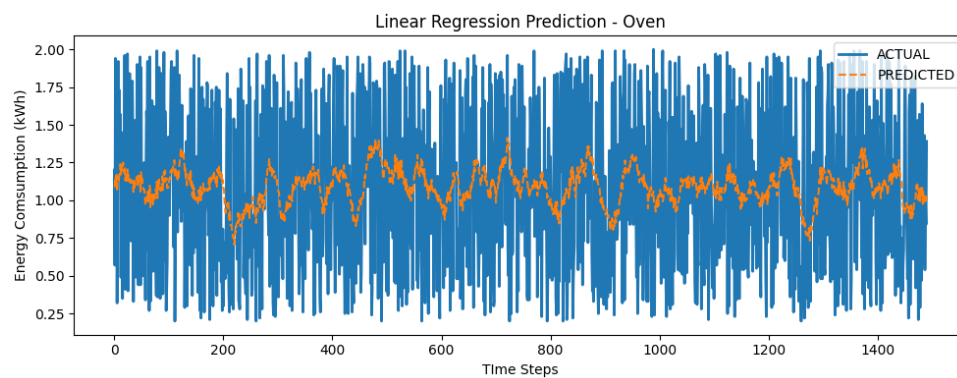
RMSE : 0.5031



Training Linear Regression for: Oven

MAE : 0.4384

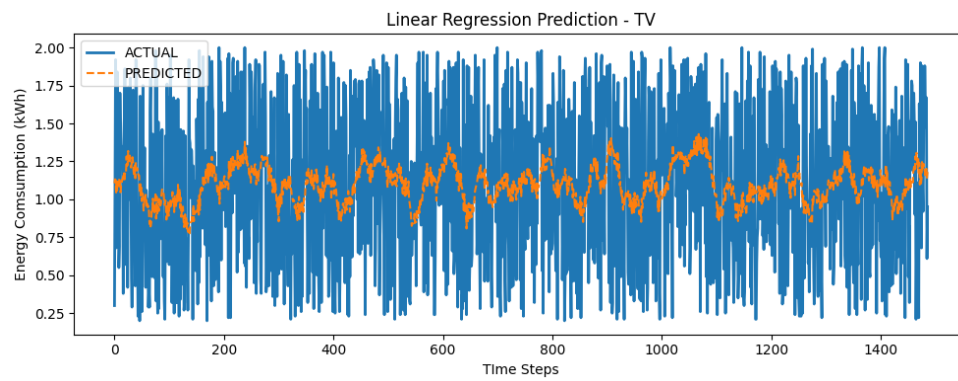
RMSE : 0.5100



Training Linear Regression for: TV

MAE : 0.4382

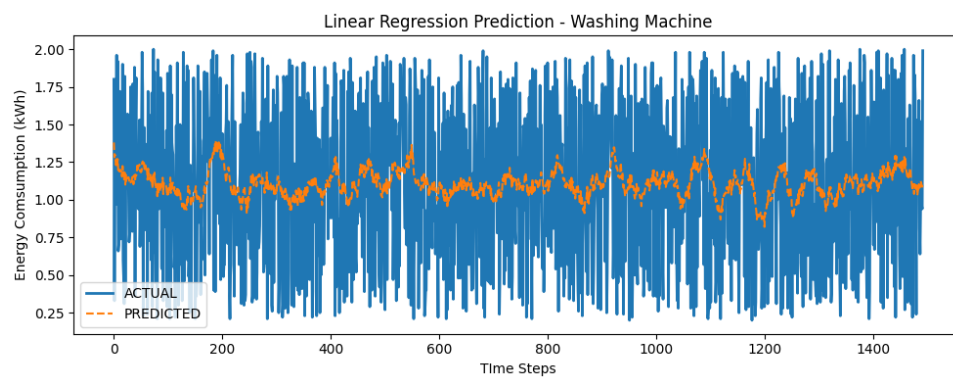
RMSE : 0.5078



Training Linear Regression for: Washing Machine

MAE : 0.4406

RMSE : 0.5091



The Error Metrics I got

Appliance	MAE	RMSE
Air Conditioning	0.683017	0.799908
Computer	0.436321	0.503666
Dishwasher	0.441374	0.513663
Fridge	0.098517	0.113672
Heater	0.695515	0.804875
Lights	0.428921	0.499550
Microwave	0.433094	0.503119
Oven	0.438439	0.509963
TV	0.438239	0.507798
Washing Machine	0.440626	0.509069